A strict upper bound for the bidisc Bohr radius: an exact asymmetric rational-inner certificate

Automated mathematical-discovery run `fa_banach_001`, lane 11

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Status. Candidate substantial partial result, likely valid, pending human review. The packet proves a new explicit upper bound, but it does not determine the exact value of the bidisc Bohr radius.

1 Source question and result

Let

$$f(z,w) = \sum_{m,n\geq 0} c_{mn} z^m w^n \in H^\infty(\mathbb{D}^2), \qquad \mathcal{B}_r(f) := \sum_{m,n\geq 0} |c_{mn}| r^{m+n}.$$

The bidisc Bohr radius is

$$K_2 := \sup\{r > 0 : \mathcal{B}_r(f) \leq 1 \text{ whenever } \|f\|_{H^\infty(\mathbb{D}^2)} \leq 1\}.$$

Knese's paper [1] proves $K_2 \geq 0.3006$ and, on PDF page 4 immediately after Corollary 1.2, records the lack of a nontrivial upper bound beyond $K_2 \leq K_1 = 1/3$:

> **Corollary 1.2.**
>
> $$K(\mathcal{A}_2) = K_2 \geq 0.3006.$$
>
> This is the best lower bound on K_2 we have seen in the literature. We do not know of any non-trivial upper bounds on K_2 beyond $K_1 = 1/3$.
>
> Next, we obtain an explicit lower bound for SA_d.
>
> **Theorem 1.3.** *For $d \geq 2$*

A later paper of Baran, Pikul, Woerdeman, and Wojtylak [2] gave $K_2 < 0.3177$. The result below improves that bound.

Theorem 1. *The two-dimensional Bohr radius satisfies*

$$\boxed{0.3006 \leq K_2 < \frac{313}{1000} = 0.313.}$$

More specifically, there is an explicit rational inner function $F \in H^\infty(\mathbb{D}^2)$ with $\|F\|_\infty = 1$ and $\mathcal{B}_{313/1000}(F) > 1$.

2 Proof intuition

The one-variable extremizers are disk automorphisms applied to the coordinate function. On the bidisc, composing an automorphism with a *symmetric* rational inner function still approaches the one-variable threshold from the safe side. The useful move is to break the symmetry. Reflect the stable linear polynomial $1 - xz - (1-x)w$ to obtain a degree-$(1,1)$ rational inner function ψ_x. At $x = 1/6$, the coefficients of the resolvent $\psi_x/(1 - a\psi_x)$ have enough cancellation before absolute values are taken that their absolute sum is unusually large. With the exact rational choice $a = 999/1000$, only the terms through total degree eight already force the Bohr sum above one at $r = 313/1000$.

Two elementary identities make the certificate rigorous. A sum-of-squares identity proves directly that ψ_x maps the bidisc to the unit disk. A four-term denominator recurrence then computes every coefficient in exact rational arithmetic; the final strict inequality is an integer comparison.

3 The Schur function

Fix $0 < x < 1$, put $y = 1 - x$, and define

$$p(z, w) = 1 - xz - yw, \qquad \widetilde{p}(z, w) = zw - xw - yz, \qquad \psi_x = \frac{\widetilde{p}}{p}. \tag{1}$$

Lemma 1. *The denominator p has no zeros in $\mathbb{D}^2$, and $|\psi_x(z, w)| < 1$ for $(z, w) \in \mathbb{D}^2$. Thus ψ_x belongs to the bidisc Schur class. Its radial boundary values have modulus one almost everywhere, so $\|\psi_x\|_\infty = 1$.*

Proof. If $|z|, |w| < 1$, then $|xz + yw| < x + y = 1$, so $p(z, w) \neq 0$. A direct expansion gives

$$|p(z, w)|^2 - |\widetilde{p}(z, w)|^2 = x(1 - |w|^2)|1 - z|^2 + y(1 - |z|^2)|1 - w|^2. \tag{2}$$

The right side is strictly positive in $\mathbb{D}^2$, proving $|\psi_x| < 1$. When $|z| = |w| = 1$, it vanishes and $|\widetilde{p}| = |p|$ wherever the quotient is defined. Radial limits therefore have modulus one almost everywhere, and the supremum norm is one. $\square$

For $0 < a < 1$, let

$$F_{a,x}(z, w) = \frac{a - \psi_x(z, w)}{1 - a\psi_x(z, w)}. \tag{3}$$

The scalar map $\zeta \mapsto (a - \zeta)/(1 - a\zeta)$ is a disk automorphism. Lemma 1 therefore shows that $F_{a,x}$ is rational inner and $\|F_{a,x}\|_\infty = 1$.

4 Exact coefficient certificate

Take from now on

$$x = \frac{1}{6}, \qquad y = \frac{5}{6}, \qquad a = \frac{999}{1000}, \qquad r = \frac{313}{1000}.$$

Write

$$Q := \frac{\psi_x}{1 - a\psi_x} = \frac{\widetilde{p}}{p - a\widetilde{p}} = \sum_{m,n \geq 0} q_{mn} z^m w^n.$$

Since $q_{00} = 0$, equation (3) becomes

$$F_{a,x} = a - (1 - a^2)Q. \tag{4}$$

The denominator of Q is

$$D(z,w) = 1 + \frac{799}{1200}z - \frac{4001}{6000}w - \frac{999}{1000}zw. \tag{5}$$

Let $n_{11} = 1$, $n_{01} = -1/6$, $n_{10} = -5/6$, and let every other n_{mn} be zero. Comparing coefficients in $DQ = \widetilde{p}$ yields, with $q_{ij} = 0$ when either index is negative,

$$q_{mn} = n_{mn} - \frac{799}{1200}q_{m-1,n} + \frac{4001}{6000}q_{m,n-1} + \frac{999}{1000}q_{m-1,n-1}. \tag{6}$$

Define the finite layer norms

$$T_k := \sum_{m+n=k} |q_{mn}|.$$

Exact application of (6) gives

k	T_k
1	1
2	21983/18000
3	18642677/12000000
4	950105595503/648000000000
5	4751192992371613/3888000000000000
6	39242182643309639173/23328000000000000000
7	2191652817952769393981/1093500000000000000000
8	1463094271925643860042526233/8398080000000000000000000000

Consequently,

$$\begin{aligned} S_8 &:= \sum_{k=1}^{8} T_k r^k \\ &= \frac{420207945895146052725713514433991908771800854450393}{8398080000000000000000000000000000000000000000000000}. \end{aligned} \tag{7}$$

The relevant threshold is $1/(1+a) = 1000/1999$, and exact subtraction gives

$$S_8 - \frac{1000}{1999} = \frac{187683844396959398701315353549825634829908046335607}{16787761920000000000000000000000000000000000000000000000} > 0. \tag{8}$$

By (4), positivity of every omitted absolute-value term, and (8),

$$\begin{aligned} \mathcal{B}_r(F_{a,x}) &\geq a + (1-a^2) \sum_{1 \leq m+n \leq 8} |q_{mn}| r^{m+n} \\ &= a + (1-a^2)S_8 > a + \frac{1-a^2}{1+a} = 1. \end{aligned}$$

Thus the defining Bohr inequality fails at $r = 313/1000$. Since the finite subsum depends continuously on r and is strictly greater than one there, it also fails on a left neighborhood of $313/1000$. Hence $K_2 < 313/1000$, proving Theorem 1. $\square$

5 Verification

The included script in the `code` directory uses only Python's standard-library exact rational-arithmetic class. It:

1. recomputes all q_{mn} with $1 \le m+n \le 8$ from recurrence (6);
2. independently checks every coefficient residual in $DQ = \widetilde{p}$;
3. checks all eight displayed T_k values;
4. checks (7), the strict threshold inequality, and the final positive Bohr-mass margin.

The command is

```
conda run --no-capture-output -n sandbox python code/verify_certificate.py
```

The verifier prints `PASS`. Because all inputs and operations are rational, this computation is an exact finite proof certificate rather than a floating-point assumption. Identity (2) was also checked by direct symbolic expansion and is short enough to audit by hand.

6 Scope, novelty check, and reviewer focus

The exact value of K_2 remains open; this theorem only narrows the interval from the best bounds found in the search to

$$0.3006 \le K_2 < 0.313.$$

On 12 August 2026, bounded searches covered the exact source title and arXiv id, the phrases "bidisc Bohr radius", "K_2 Bohr radius polydisk", "non-trivial upper bounds on K_2", and "asymmetric rational inner Bohr radius bidisk", together with the citations and later papers visible from the source. The separate post-source improvement found was Baran–Pikul–Woerdeman–Wojtylak [2, Theorem 6.4], which proves $K_2 < 0.3177$ using a different numerically optimized degree-$(1,1)$ rational inner function and an exact degree-twelve coefficient check. No searched source states $K_2 < 0.313$ or uses the one-parameter family (1). The novelty assessment is therefore *plausible, not certified*; a specialist should conduct a broader bibliographic review.

The most useful human checks are: (i) expand the elementary identity (2); (ii) verify the sign convention in recurrence (6); and (iii) rerun the exact certificate. No claim depends on the numerical scan that suggested $x = 1/6$.

References

[1] G. Knese, *Three radii associated to Schur functions on the polydisk*, arXiv:2410.21693v3 (2025).

[2] R. Baran, P. Pikul, H. J. Woerdeman, and M. Wojtylak, *Contractive realization theory for the annulus and other intersections of discs on the Riemann sphere*, arXiv:2504.03236 (2025), Theorem 6.4.